

Muni Madhuri Koilakuntla

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PROFESSIONAL SUMMARY

- Java Full Stack Developer with **5+ years of experience** designing, developing, and supporting scalable, cloud-native enterprise applications across Financial Services, Payments, Healthcare, and eCommerce domains.
- Strong expertise in **Java, Spring Boot, Microservices Architecture, RESTful APIs, SQL, Hibernate, JPA, and Full Stack Development** using **frontend** frameworks like **React, Angular, TypeScript, JavaScript, HTML5, and CSS3**.
- Experienced in **DevOps** automation with tools such as **Jenkins, Git, GitLab, CI/CD pipelines, Maven, Gradle, Terraform, and Ansible**, enabling consistent **production-ready systems** and workflow optimization.
- Strong expertise in **Google Cloud Platform (GCP), Azure** and **AWS** services including **Lambda, EKS, S3, SNS, and SQS**, along with event-driven architectures leveraging **Apache Kafka, JMS** and **ActiveMQ**.
- Familiar with databases including **SQL, MySQL, PostgreSQL, MongoDB, DynamoDB, Oracle** and **Microsoft SQL Server**, caching using **Redis** and **NoSQL** and knowledgeable in containerization with **Docker, Kubernetes, and AWS deployments**.
- Proficient in unit testing and integration testing automation using **JUnit, Mockito, Selenium, and Playwright** with strong debugging, performance tuning, and deployment experience.
- Adept at security implementations using **OAuth 2.0, JWT, and Spring Security**, and proficient in monitoring tools like **ELK stack, Prometheus, and CloudWatch** for error handling and system tracking.
- Extensive experience working in **Agile and Scrum environments**, practicing **Test-Driven Development (TDD)** and **Behavior Driven Development (BDD)**, while mentoring junior developers, conducting code reviews, and promoting engineering best practices.

WORK EXPERIENCE

Software Engineer

Jan 2025 – Present

Mastercard, Chicago, Illinois, USA

- Design, develop, and maintain scalable web applications using **Java 8+**, **Spring Boot** microservices, and **RESTful APIs** supporting **high-volume payment systems** ensuring scalability, reliability, and security.
- Build **responsive front-end applications** using **React, JavaScript, TypeScript, HTML, and CSS**, improving user experience and reducing UI latency by **30%**.
- Deployed cloud-native services on **AWS**, including **EC2, S3, and Lambda**, using **Docker and Kubernetes**, integrating CI/CD pipelines with **Jenkins and GitHub Actions**, enabling **15+ production releases per month**.
- Optimized **SQL queries, ORM mappings, relational database** using **Hibernate, JPA, indexing strategies, and transactional workflows** to support high-throughput transaction systems, ensuring data integrity and reliable processing across microservices.
- Enhanced data processing workflows with **Apache Spark, Apache Kafka, and RabbitMQ**, ensuring efficient real-time and batch data handling, improving performance by **25%**.
- Implemented monitoring and testing solutions with **Prometheus, Grafana, JUnit, and Mockito**, ensuring high application reliability and reducing production issues.
- Implemented monitoring solutions using **Ansible and OpenShift**, and managed backups, repairs, and alerts for **Apache Cassandra** clusters with **OpsCenter**.
- Collaborate with cross-functional teams, Product Owners, Business Analysts, Stakeholders, QA, and DevOps teams in **Agile and Scrum sprints**, participating in sprint planning, code reviews, and feature demos.
- Applied Data Structures and algorithms using **core Java** and **Object-Oriented Programming (OOP)** principles including Collections framework like Lists, Set, Map, exception handling and multithreading concepts to optimize application logic and improve performance.
- Followed **Agile methodologies** like **TDD**, developing comprehensive unit tests with **JUnit** for reliable software quality coverage.

Software Engineer

May 2024 – Dec 2024

Abbott Laboratories, Chicago, Illinois,, USA

- Developed modular microservices in **Spring Boot** and **Java**, ensuring secure API access using **OAuth 2.0, JWT, and SAML 2.0** protocols. Exposed endpoints via **REST** and **SOAP** web services using **CXF API**.
- Built high-performing front-end components using **React.js, JavaScript, Backbone.js, HTML5, and CSS3**, ensuring cross-browser compatibility and responsive, user-friendly UI interfaces for real-time data visualization.
- Implemented **real-time communication** using **WebSocket** and designed **GraphQL APIs** for efficient data querying and flexible front-end integration.
- Leveraged **Apache Kafka** for building real-time streaming pipelines, integrated **Redis** for caching frequently accessed data, and utilized **Apache Flink** for processing high-volume data streams.
- Automated deployment workflows with **Terraform** and **Ansible**, utilized **Docker** for containerization, and orchestrated applications using **Kubernetes** to improve scalability and fault tolerance.

- Optimized performance monitoring using **Splunk**, **Prometheus**, and **Elasticsearch**, ensuring proactive troubleshooting and advanced search capabilities for large-scale systems.
- Created unit and integration tests with **JUnit**, **SOAPUI** and **Log4J** to validate services, reliability, trace logs and identify root causes.
- Managed database operations using **MySQL**, **Oracle**, and **Spring Data JPA**, supporting transactional and analytical use cases, including legacy system data migration.
- Implemented authentication and authorization protocols using **OAuth 2.0**, **JWT**, and **Spring Security**, maintaining compliance with internal access control policies.
- Developed serverless solutions using **AWS Lambda** for cost-effective event-driven computing and utilized **Swagger** for comprehensive API documentation to streamline collaboration with front-end teams.
- Implemented **Redis** and **Elasticsearch** for log management and caching, optimizing data access times and streamlining application debugging. Enabled agile collaboration using **JIRA** and **Git** for user story management, sprint reviews and version controls.
- Collaborated with QA teams to write and maintain unit and integration tests using **JUnit**, **Mockito**, and **Selenium**, achieving over **90% test coverage with SonarQube** and minimizing bugs in production.
- Developed real-time data streaming solutions for patient monitoring systems by integrating **Java** with **Kafka**, ensuring compliance with healthcare standards such as **HL7** and **FHIR**.

Senior Software Engineer

Jun 2021 - Jan 2024

Capgemini Technology Services, Hyderabad, India

- Designed, developed, and maintained enterprise-grade full-stack applications for **financial** services and **insurance** platforms using **Java**, **Spring MVC**, **Rest APIs**, and **microservices architecture** with **Spring Cloud** deployed on **Pivotal Cloud Foundry (PCF)**.
- Modernized front-end with **Angular**, **React**, **TypeScript**, **JavaScript**, **HTML5**, **CSS3** and **Bootstrap**, delivering responsive, modular UI components via **AJAX**, **JSON** that improved client satisfaction by reducing ticket escalations in production environment.
- Implemented event-driven architectures between distributed systems using **Kafka**, **Databricks**, **DataLake**, and **Apache NiFi pipelines** to process high-volume financial transactions and ingestion workflows.
- Developed and deployed **Spring Boot** and **Node.js microservices** using **NetBeans IDE**, integrated with **WebLogic servers**, and enhanced application monitoring with **Log4J** and **Splunk**.
- Built & consumed **SOAP** and **RESTful APIs**, **web services** compatible with **XML** and **JSON** payloads enabling seamless **integration framework** with third-party services and external vendors for real-time insurance policy updates.
- Automated deployment containerization using **Terraform**, **Docker**, **Kubernetes**, and **Spring Cloud**, enabling zero-downtime releases and accelerating release cycles, fault tolerance and reducing deployment failures.
- Integrated **MySQL** and **Hibernate** for database design, optimization and management to maintain data integrity and security, implemented caching using **Redis**, ensured **ACID** compliance, **CRUD** operations and improved query performance.
- Leveraged **AWS cloud services (EC2, S3, SNS, Lambda, RDS)**, and **Azure DevOps** for scalable deployments and participated in data warehouse integration (**Snowflake**) for analytics and performance reporting.
- Ensured robust unit and integration testing and **CI/CD** automation using **JUnit**, **Mockito**, **Jenkins**, and **SonarQube**, used **JASPER Reports**, maintaining consistent code quality for automated deployment, testing, and version control.
- Provided **L3 production support** by troubleshooting issues, analyzing logs, and performing root cause analysis, applied code reviews, refactoring, and design patterns to resolve **40+** high-priority production issues.
- Collaborated with **cross-functional teams** including Portfolio Managers, Product Owners, Business Analysts, and stakeholders in **Agile Sprint ceremonies** to gather requirements, contributing to sprint planning, daily **scrum** meetings to deliver high quality software solutions.
- Well-versed in **SDLC**, **Agile methodology**, Version control (**Git**), Defect Tracking tool (**Jira**).

Software Engineer – Java Developer

Jun 2019 – Dec 2019

WTA Consulting, Hyderabad, India

- Contributed to the development of **microservice**-based backend modules using **Core Java** and **Spring Boot**, integrating **REST APIs** and **JWT based authentication** for secure communication between client and server components.
- Implemented **CI/CD** automation pipelines with **Azure DevOps**, **Docker**, and automated **ETL batch processes** using **Autosys** and **UNIX Shell scripting**, integrated with **SQL invoker**.
- Monitored application health and optimized query performance in **Apache Cassandra** based distributed systems using **Log4J**, **SLF4J**, and **OpsCenter**, reducing system downtime and latency.
- Worked with **JSP**, **J2EE**, and **JDBC** to develop server-side components, handle database connectivity, and support enterprise Java application workflows.
- Assisted QA efforts by writing unit test cases and basic automation scripts achieving **85%** coverage, contributed to build stability and reduced manual rework through test-driven development (**TDD**) practices in integration cycles.
- Deployed and managed Java applications on **Apache Tomcat** and **RabbitMQ**, ensuring smooth application startup, configuration, and runtime stability and managed source using **Git** version control.

SKILLS

- **Programming Languages:** Java, Python, SQL, JavaScript, TypeScript, Go, Shell Script, Bash, HTML5, CSS
- **Frontend Frameworks:** Angular, React, Next.js, Vue.js, Bootstrap, Thymeleaf, XML, JSON, SOAPUI
- **Backend Frameworks:** J2EE, Node.js, Spring Boot, Spring Framework, Spring MVC, Microservices Architecture, RESTful APIs, SOAP Web Services, API Development(REST, JSON), Hibernate, ORM, JPA, JWT, JMS, MQ
- **Databases:** SQL, NoSQL, MySQL, PostgreSQL, MongoDB, Oracle, DynamoDB, PL/SQL, DataLake, ADX, Databricks
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Microsoft Azure DevOps, Google Cloud Platform (GCP), Docker, Kubernetes, OpenShift, Artifactory, Jenkins, CI/CD, GitLab, GitHub Actions
- **Infrastructure & Monitoring:** Terraform, AWS CloudFormation, ELK Stack, Prometheus, Grafana, Dynatrace, Splunk, Fluentbit
- **Messaging & Streaming:** Apache Kafka, JMS, RabbitMQ, ActiveMQ, Apache NiFi, Tomcat
- **MLOps & ML Tools:** Scikit-learn, TensorFlow, Pandas, SageMaker, MLflow, JUnit, SonarQube
- **Tools & Version Control:** Spring Boot, VS Code, Eclipse, Guidewire, Maven, Gradle, Git, GitHub, Bitbucket, Confluence
- **Testing:** JUnit5, Mockito, Selenium, Cucumber, Playwright, Karma, Jasmine, Cypress, Postman, Swagger, TestNG
- **Methodologies:** Agile, Scrum, SDLC, Test-Driven Development (TDD), DevSecOps, Behavior Driven Development(BDD)

EDUCATION

- **Master of Science in Computer Science**
Southern Illinois University Carbondale, IL, USA